

PAPER_411 – Ug1: Di-Pseudo-Monopole (DPM) Internal Dipole — Solar Calibration and Defect Mechanism

Session: 110 (grokshare755feea7.txt analysis)

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Source: grokshare755feea7.txt — "Star Magic" Chapter 1, Section Ug1, Solar Refinement **Session:** 110 (grokshare755feea7.txt analysis) **CP4 Class:** Ug1DPMDiPseudoMonopoleSolarCalibrationCalculator (#61)

1. Overview

PAPER_411 establishes **Ug1 — the Di-Pseudo-Monopole (DPM)** as the foundational internal driving force of any star. Ug1 is the **first and primary** gravity range in the UQFF hierarchy, capturing the stellar dipole moment arising from the interaction of Universal Aether derivatives (UA') with trapped SCm.

Key contributions of Ug1:

Drives surface irregularities through internal defects (δ_{def})

Is the **origin force** from which Ug2, Ug3, Ug4, and Ug4i all cascade

Is modulated by π cycles and non-linear time decay

Has been calibrated using specific solar data: $\mu_s \approx 3.38 \times 10^{20} \text{ T}\cdot\text{m}^3$ (base)

2. Ug1 Equation

$$Ug_1 = k_1 \cdot \mu_s(t, \rho_{\text{vac}}, [\text{SCm}]) \cdot \nabla \left(\frac{M_s}{r} \right) e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + \delta_{\text{def}})$$

where:

Symbol	Value (Sun)	Description
k_1	1.5	Coupling constant for Ug1 (refined)
μ_s	$(10^3 + 0.4 \sin(\omega_c t)) \cdot 3.38 \times 10^{20} \text{ T}\cdot\text{m}^3$	Stellar DPM moment with SCm
$\nabla(M_s/r)$	$\approx 274 \text{ m/s}^2$	Gradient of gravitational potential at $r = R_s$
α	0.001 day^{-1}	Non-linear time decay rate
$\cos(\pi t_n)$	oscillatory	π cycle modulation with negative time
δ_{def}	$0.01 \cdot \sin(0.001 \cdot t)$	Defect factor — drives surface irregularities

3. Di-Pseudo-Monopole Physics

3.1 DPM Definition

The Di-Pseudo-Monopole is identified as:

$$\text{DPM} \equiv \frac{[UA']}{[\text{SCm}]}$$

where $[UA']$ is the first-order derivative of the Universal Aether, meaning the **gradient flux** of Aether as it interacts with the internal SCm configuration. The monopole character arises because adjacent field lines cannot escape (monopole-like), but the system is an internal dipole (hence "di-pseudo").

3.2 Stellar DPM Moment (μ_s)

For a star, the DPM moment is:

$$\mu_s(t, \text{SCm}) = \left[B_s(t) + B_{\text{SCm}} \right] \cdot R_s^3$$

where:

$$B_s(t) = B_s^{(0)} + 0.4 \cdot \sin(\omega_c t) \text{ — time-varying surface field}$$

$$B_{\text{SCm}} \approx 10^3 \text{ T — SCm-driven superconductive contribution (undetectable via } Q_s = 0)$$

Solar Values:

$$B_s^{(0)} = 10^{-4} \text{ T}, \quad R_s = 6.96 \times 10^8 \text{ m}$$

Base DPM moment (no SCm):

$$\mu_{s,\text{base}} = 10^{-4} \cdot (6.96 \times 10^8)^3 \approx 3.38 \times 10^{20} \text{ T}\cdot\text{m}^3$$

Full DPM moment (with SCm):

$$\mu_{s,\text{full}} \approx (10^3 + 0.4 \sin(\omega_c t)) \cdot 3.38 \times 10^{20} \\ \approx 3.38 \times 10^{23} + 1.35 \times 10^{20} \sin(\omega_c t) \text{ T}\cdot\text{m}^3$$

The **SCm contribution dominates** by 7 orders of magnitude over the bare magnetic field.

4. Gravitational Gradient Term

$$\nabla \left(\frac{M_s}{r} \right) \approx \frac{G M_s}{R_s^2} = \frac{6.674 \times 10^{-11} \cdot 1.989 \times 10^{30}}{(6.96 \times 10^8)^2} \approx 274 \text{ m/s}^2$$

This is the **surface gravity** of the Sun, confirming dimensional alignment of the DPM gradient term.

5. Numerical Calibration at $t = 0$

With $k_1 = 1.5, \Delta_{\text{def}} = 0, \cos(\pi \cdot 0) = 1$:

$$U_g(t=0) \approx 1.5 \cdot 3.38 \times 10^{23} \cdot 274 \cdot 1 = 1.39 \times 10^{26} \text{ [normalized units]}$$

With solar cycle oscillation:

$$Ug_1(t) \approx (1.39 \times 10^{26} + 5.55 \times 10^{23} \cdot \sin(\omega_c t)) \cdot e^{-0.001t} \cdot \cos(\pi t)$$

6. Surface Irregularities via δ_{def}

The defect factor models surface phenomena (sunspots, flares, magnetic anomalies):

$$\delta_{\text{def}}(t) = 0.01 \cdot \sin(0.001 \cdot t)$$

This modifies Ug_1 by $\pm 1\%$ with a period of roughly 6,280 seconds (~ 1.7 hours), representing **rapid surface defect cycles** driven by the internal SCm structure.

The Sun's observable surface activity (sunspots, differential rotation patterns) is a **direct readout** of these Ug_1 internal defects — the surface magnetism is **unique to the surface**, not the interior.

7. Cascade to Ug_2 , Ug_3 , Ug_4

Ug_1 is the generative force for all higher Ug terms:

Cascade Effect	Mechanism
$Ug_1 \rightarrow Ug_2$	DPM field bubble expands outward, forming the heliosphere via charge-reactive Aether trapping
$Ug_1 \rightarrow Ug_3$	Equatorial CCW vs coronal CW spin differential (arising from DPM asymmetry) creates the magnetic strings disk
$Ug_1 \rightarrow Ug_4$	DPM field propagates to the star–black hole interaction scale via vacuum energy modulation
$Ug_1 \rightarrow Ug_{4i}$	Sub-range DPM effects extend to galactic vacuum fluctuation coupling

8. Code Implementation

```
double compute_mu_s(double t, double Bs, double omega_c, double Rs, double SCm_contrib = 1e3) {
    double Bs_t = Bs + 0.4 * std::sin(omega_c * t) + SCm_contrib;
    return Bs_t * std::pow(Rs, 3);
}

double compute_grad_Ms_r(double Ms, double Rs) {
    if (Rs <= 0.0) throw std::runtime_error("Invalid Rs value");
    return G * Ms / (Rs * Rs);
}

double compute_Ug1(const CelestialBody& body, double r, double t, double tn,
    double alpha, double delta_def, double k1) {
    if (r <= 0.0) throw std::runtime_error("Invalid r value");
    double mu_s = compute_mu_s(t, body.Bs_avg, body.omega_c, body.Rs);
    double grad_Ms_r = compute_grad_Ms_r(body.Ms, body.Rs);
    double defect = 1.0 + delta_def * std::sin(0.001 * t);
    return k1 * mu_s * grad_Ms_r * std::exp(-alpha * t) * std::cos(PI * tn) * defect;
}
```

9. Unit Test

```
def test_Ug1_solar_calibration():
    """Solar Ug1 at t=0, no defect"""
    import math
    G = 6.674e-11; Ms = 1.989e30; Rs = 6.96e8
    Bs = 1e-4; SCm_contrib = 1e3; k1 = 1.5; alpha = 0.001; t = 0; tn = 0
    mu_s = (Bs + SCm_contrib) * Rs**3 # ≈ 3.38e23
    grad_Ms_r = G * Ms / Rs**2 # ≈ 274
    defect = 1.0
    expected = k1 * mu_s * grad_Ms_r * math.exp(-alpha * t) * math.cos(math.pi * tn) * defect
    # ≈ 1.39e26
    assert expected > 1e25, f"Ug1 solar calibration failed: {expected}"
```

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